Singular Value Decomposition: Rotate, Stretch, Rotate

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Linear transformations and matrices; Eigenvectors and eigenvalues.
- **Leads to:** Low-rank approximation (truncated SVD); PCA; Latent semantic analysis; Moore-Penrose pseudo-inverse.
- **Compare with:** Eigenvectors and eigenvalues.

1. Overview

Key point: Every matrix, square or not, can be written as $A = U\Sigma V^\mathsf{T}$: a **rotation** (G-1709), then a stretch along the axes, then another rotation. The stretch factors are the singular values.

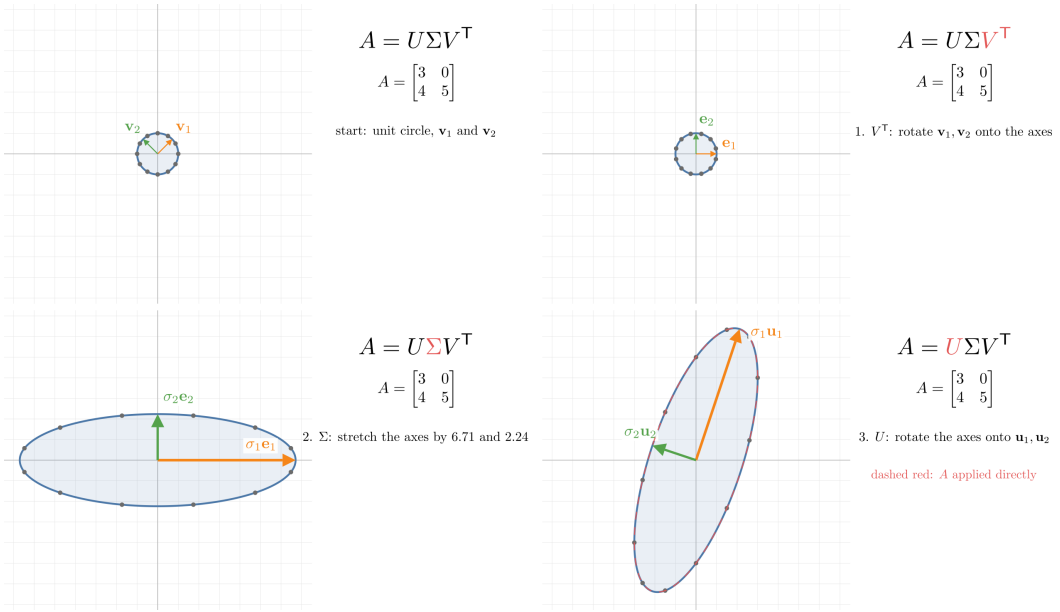


Figure 1: The SVD of A with rows $[3, 0]$ and $[4, 5]$ applied to the unit circle one factor at a time: V^T rotates, Σ stretches, U rotates

The **eigenbasis** (G-664) lets us write some square matrices as $A = PDP^{-1}$, with D diagonal. That eigen-decomposition works only for square matrices with enough eigenvectors. The **singular value decomposition** (G-1813) (**SVD**) is a factorisation that works for every matrix. The linear algebra roadmap lists it among the factorisations ML uses.

This Note builds it from the picture first:

- the matrices that only rotate or flip, called orthogonal matrices (Section 2);
- every matrix turns the unit circle into an ellipse (Section 3);

- $A = U\Sigma V^{\mathsf{T}}$ as rotate, stretch, rotate (Section 4 and Figure 1);
- **singular values** (G-1812) and singular vectors (Section 5);
- matrices that are not square, and the **thin SVD** (G-1969) (Section 6);
- how the SVD differs from the eigen-decomposition (Section 7).

How to compute the factors by hand comes later in computing the SVD.

One matrix runs through the whole Note:

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

Read as a transformation (as in the matrix of a transformation), A sends $\hat{\imath}$ to $[3, 4]$ and $\hat{\jmath}$ to $[0, 5]$.

2. Orthogonal matrices: moves that only rotate or flip

> **Key point:** A square matrix whose columns are perpendicular unit vectors is an **orthogonal matrix** (G-1407). It only rotates or flips space, keeps every length and angle, and its inverse is its transpose.

2.1 Recap: a matrix moves the grid

A matrix moves every point of the plane so that grid lines stay straight, parallel and evenly spaced, and the origin stays put: a **linear transformation** (G-1097). Its first column is where $\hat{\imath}$ lands and its second column is where $\hat{\jmath}$ lands. Reading a matrix as a picture teaches this picture; the SVD is built entirely from it.

2.2 Moves that keep every length and angle

> **Key point:** Some matrices turn the plane without stretching it: every arrow keeps its length, and every pair of arrows keeps its angle.

Draw two arrows with an angle between them, and let a matrix move the plane. Most matrices stretch the arrows and bend the angle. A few only turn the plane around, or turn it over, so that the arrows come out with the same lengths and the same angle as before: nothing is distorted.

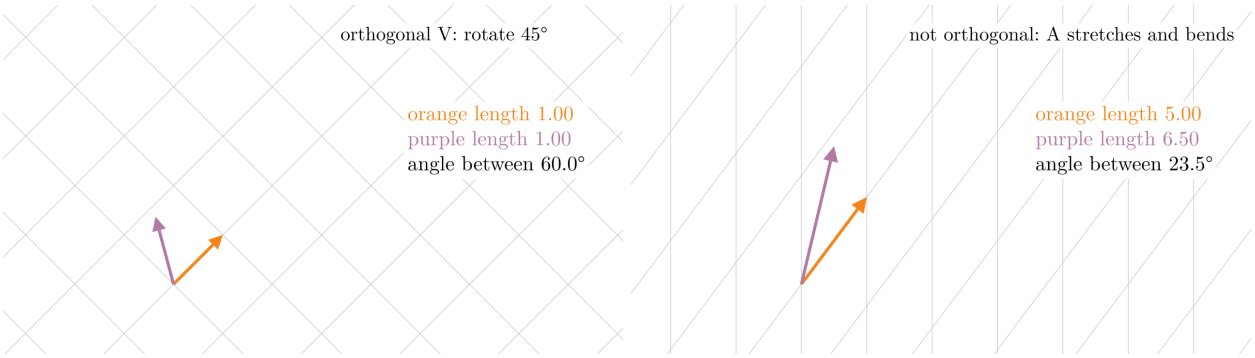


Figure 2: Two unit arrows, 60° apart, ride along as the grid moves. Under the rotation V (45°) their lengths stay 1 and their angle stays 60°; under A (rows $[3, 0]$, $[4, 5]$) they grow to 5 and 6.50 and close to 23.5°. Idea after Khan Academy, "Orthogonal matrices preserve angles and lengths"

In Figure 2, watch the three readouts. During the turn they never change. During A all three change: A is not one of these distortion-free moves.

2.3 Orthonormal columns

> **Key point:** Unit vectors that are all perpendicular to each other are called orthonormal. A matrix whose columns are orthonormal is an orthogonal matrix.

A set of vectors is **orthonormal** (G-1409) when each has length 1 and each pair is perpendicular (**orthogonal**, G-1408). A square matrix whose columns are orthonormal is an **orthogonal matrix** (G-1407). Our first example is

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Its columns $[0.707, 0.707]$ and $[-0.707, 0.707]$ each have length 1:

$$\sqrt{0.5 + 0.5} = 1$$

Their dot product is 0:

$$-0.5 + 0.5 = 0$$

Section 3 will meet these two columns again as the special input directions of A, and will build a second orthogonal matrix, U, from the special output directions.

("Orthonormal matrix" would be more accurate, but "orthogonal matrix" is the convention; MML Def. 3.8.)

2.4 What an orthogonal matrix does to space

> **Key point:** Its columns are where $\hat{\imath}$ and $\hat{\jmath}$ land; they stay perpendicular unit vectors, so the grid is only turned or mirrored.

Reading the columns as landing spots of $\hat{\imath}$ and $\hat{\jmath}$, an **orthogonal matrix** (G-1407) moves the two basis vectors to two other perpendicular unit vectors. The grid squares stay squares of the same size. So nothing is stretched or slanted; space is only rotated, or rotated and mirrored.

- V sends $\hat{\imath}$ to $[0.707, 0.707]$: a **rotation** (G-1709) by 45° counterclockwise.
- U of Section 3 sends $\hat{\imath}$ to $[0.316, 0.949]$: a rotation by 71.6°, since $\tan 71.6° = 3$.
- A matrix with columns $[1, 0]$ and $[0, -1]$ is also orthogonal: a **reflection** (G-1651) that mirrors the plane across the x-axis.

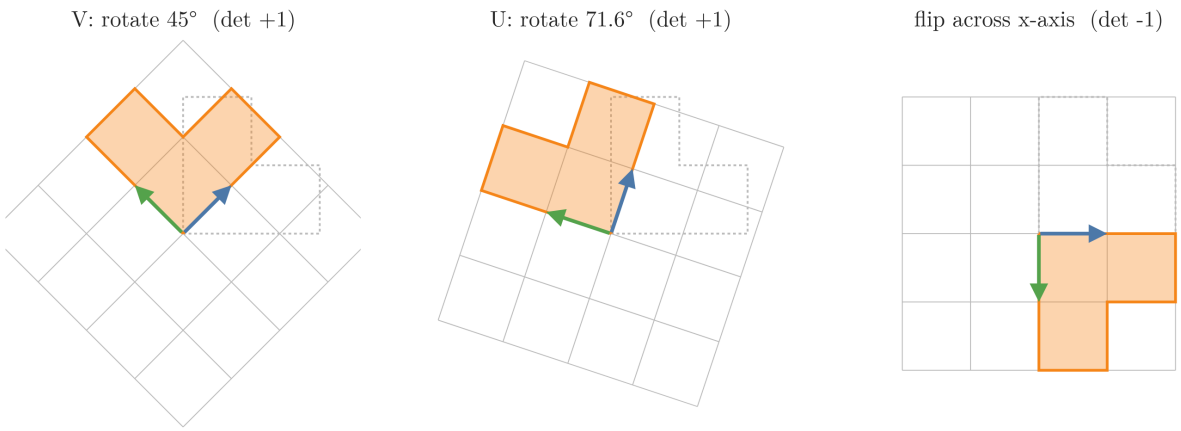


Figure 3: The three orthogonal matrices of this Note acting on a grid and an L-shaped tile (dotted: before; orange: after). Arrows: where $\hat{\imath}$ (blue) and $\hat{\jmath}$ (green) land

In Figure 3, watch the grid squares: each one stays a unit square, so lengths and angles survive. Only the flip turns the L into its mirror image. The determinant tells the two kinds apart: $+1$ for a rotation, -1 for a rotation combined with a flip. The determinant is never anything else, because areas do not change.

2.5 Why lengths and angles survive: the inverse is the transpose

> **Key point:** For an **orthogonal matrix** (G-1407) Q, $Q^{\mathsf{T}}Q = I$. That one fact keeps every length and angle, and makes undoing Q free: $Q^{-1} = Q^{\mathsf{T}}$.

1. **In words:** entry (i, j) of $Q^{\mathsf{T}}Q$ is the dot product of column i with column j. The dot product is 1 when $i = j$ (unit length) and 0 otherwise (perpendicular), so the product is the **identity matrix** (G-915).

2. **Formula:**

$$Q^{\mathsf{T}}Q = I$$

$$Q^{-1} = Q^{\mathsf{T}}$$

3. **Example:**

$$V^{\mathsf{T}}V = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$V^{\mathsf{T}}V = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$V^{\mathsf{T}}V = I$$

So V^{T} is the **rotation** (G-1709) by $-45°$, the move that undoes V.

4. **Why lengths survive:** a squared length is a vector's dot product with itself, $\|\mathbf{x}\|^2 = \mathbf{x}^{\mathsf{T}}\mathbf{x}$. So

$$\begin{aligned} \|Q\mathbf{x}\|^2 &= (Q\mathbf{x})^{\mathsf{T}}(Q\mathbf{x}) \\ &= \mathbf{x}^{\mathsf{T}}Q^{\mathsf{T}}Q\,\mathbf{x} \\ &= \mathbf{x}^{\mathsf{T}}\mathbf{x} \\ &= \|\mathbf{x}\|^2 \end{aligned}$$

The step from the second line to the third uses $Q^{\mathsf{T}}Q = I$. In the same way $(Q\mathbf{x})^{\mathsf{T}}(Q\mathbf{y}) = \mathbf{x}^{\mathsf{T}}\mathbf{y}$: dot products survive, and with them the cosine of every angle. This is what Figure 2 showed.

The **transpose** (G-2012) and the identity matrix were defined in projecting one point and a matrix is a transformation, and the **inverse matrix** (G-968, the matrix that undoes another) in the normal equation. Computing an inverse is normally expensive; for an orthogonal matrix it is free.

> **Extra:** Where ML uses eigenvectors says a covariance matrix (see the covariance matrix) always has perpendicular eigenvectors. Scaled to length 1 and placed as columns, they form an orthogonal matrix. The orthogonal eigenvector matrix is why PCA's projection onto the principal components is a pure rotation of the data, followed by dropping some coordinates.

3. Every matrix turns a circle into an ellipse

> **Key point:** A matrix sends the unit circle to an ellipse. There is always one perpendicular pair of input directions that lands exactly on the ellipse's two perpendicular axes.

3.1 Perpendicular in, usually not perpendicular out

> **Key point:** Most perpendicular pairs of vectors are not perpendicular any more after the matrix acts.

Take two perpendicular vectors and apply A. Usually the results are not perpendicular. The standard basis vectors are an example: $\hat{\imath}$ and $\hat{\jmath}$ meet at 90°, but they land on $[3, 4]$ and $[0, 5]$. Their dot product is:

$$3 \times 0 + 4 \times 5 = 20$$

This is not 0, so they are no longer at right angles (see perpendicular vectors have dot product 0).

3.2 The one pair that stays perpendicular

> **Key point:** For A, the unit vectors at 45° and 135° land on perpendicular vectors of lengths 6.71 and 2.24.

Turn the perpendicular pair around and at some angle the outputs become perpendicular too. Figure 4 turns the pair in 5° steps.

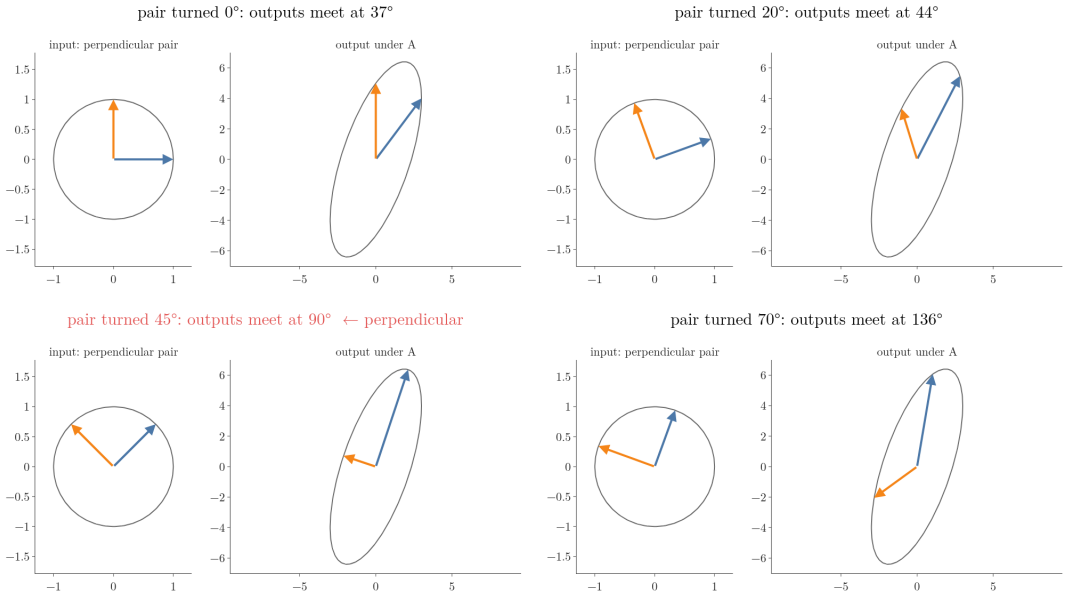


Figure 4: A perpendicular pair of unit vectors (left) turns in 5° steps; A sends it to the pair on the right. The title gives the angle between the two outputs

Watch the angle in the title: it starts at 37° for $\hat{\imath}, \hat{\jmath}$, reaches exactly 90° only at a turn of 45°, and passes it on the way to 136°. For A the perpendicular pair is

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$$
$$\mathbf{v}_1 \approx \begin{bmatrix}0.707\\0.707\end{bmatrix}$$

$$\mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}-1\\1\end{bmatrix}$$
$$\mathbf{v}_2 \approx \begin{bmatrix}-0.707\\0.707\end{bmatrix}$$

Both are unit vectors (length 1), and they are perpendicular. Their outputs:

$$A\mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}3\\9\end{bmatrix}$$
$$A\mathbf{v}_1 \approx \begin{bmatrix}2.121\\6.364\end{bmatrix}$$

$$A\mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}-3\\1\end{bmatrix}$$
$$A\mathbf{v}_2 \approx \begin{bmatrix}-2.121\\0.707\end{bmatrix}$$

The dot product of the outputs is 0, so they are still perpendicular:

$$\tfrac{1}{2}(3 \times (-3) + 9 \times 1) = 0$$

Their lengths are:

$$\sqrt{90/2} = \sqrt{45} \approx 6.71$$
$$\sqrt{10/2} = \sqrt{5} \approx 2.24$$

3.3 The ellipse

> **Key point:** The unit circle becomes an ellipse whose long axis is $A\mathbf{v}_1$ and whose short axis is $A\mathbf{v}_2$.

The **unit circle** is the set of all points at distance 1 from the origin, so every point of it is a unit vector $\mathbf{x}$. Applying A to all of them gives a closed curve, and for a **linear transformation** (G-1097) that curve is always an ellipse (Figure 5, right). Its two axes are perpendicular, and they are exactly $A\mathbf{v}_1$ and $A\mathbf{v}_2$.

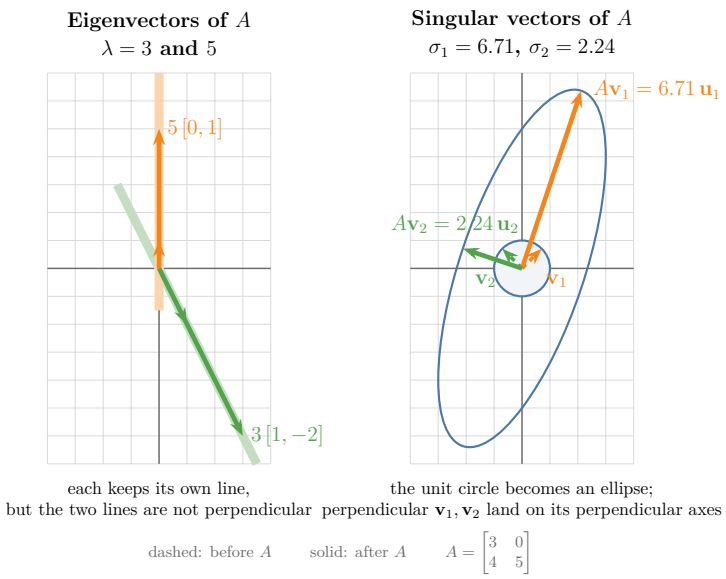


Figure 5: Eigenvectors keep their line; singular vectors are the perpendicular pair that stays perpendicular

So the longest output any unit vector can reach is 6.71, along $A\mathbf{v}_1$, and the shortest is 2.24, along $A\mathbf{v}_2$. These two numbers are the **singular values** (G-1812) of A, written $\sigma_1 = 6.71$ and $\sigma_2 = 2.24$ (σ is the Greek letter sigma).

Dividing each output by its length gives the directions of the ellipse's axes, as unit vectors:

$$\mathbf{u}_1 = \frac{A\mathbf{v}_1}{\sigma_1}$$

$$\mathbf{u}_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\mathbf{u}_1 \approx \begin{bmatrix} 0.316 \\ 0.949 \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{A\mathbf{v}_2}{\sigma_2}$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$\mathbf{u}_2 \approx \begin{bmatrix} -0.949 \\ 0.316 \end{bmatrix}$$

3.4 The singular value equation

> **Key point:** $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$: each input direction lands on its output direction, stretched by its **singular value** (G-1812). This is the **singular value equation** (G-1814).

1. **In words:** A takes the unit vector $\mathbf{v}_i$ to the unit vector $\mathbf{u}_i$, stretched by σ_i. The $\mathbf{v}$'s are perpendicular, and so are the $\mathbf{u}$'s.
2. **Formula:**
$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i$$
3. **Example:** for the first pair:
$$A\mathbf{v}_1 = [2.121, 6.364]$$
$$A\mathbf{v}_1 = 6.71 \times [0.316, 0.949] = \sigma_1 \mathbf{u}_1$$
For the second pair:
$$A\mathbf{v}_2 = [-2.121, 0.707]$$
$$A\mathbf{v}_2 = 2.24 \times [-0.949, 0.316] = \sigma_2 \mathbf{u}_2$$

The singular value equation looks like the eigenvector equation $A\mathbf{v} = \lambda\mathbf{v}$, with one difference: the vector on the right is a different vector, $\mathbf{u}_i$ instead of $\mathbf{v}_i$. Allowing the output direction to differ from the input direction is what makes this work for every matrix.

The $\mathbf{v}$'s are **orthonormal** (G-1409), and so are the $\mathbf{u}$'s. Placed side by side as columns, the $\mathbf{v}$'s form the **rotation** (G-1709) V of Section 2, and the $\mathbf{u}$'s form a second **orthogonal matrix** (G-1407), the rotation by 71.6°:

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$U = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$$

4. Rotate, stretch, rotate: $A = U\Sigma V^{\mathsf{T}}$

> **Key point:** V^{T} rotates $\mathbf{v}_1, \mathbf{v}_2$ onto the axes, Σ stretches the axes by the **singular values** (G-1812), and U rotates the axes onto $\mathbf{u}_1, \mathbf{u}_2$.

4.1 From the singular value equation to the factorisation

> **Key point:** Stack the equations $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ side by side as $AV = U\Sigma$, then multiply by V^{T}.

The two equations $A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1$ and $A\mathbf{v}_2 = \sigma_2 \mathbf{u}_2$ can be written as one matrix equation. On the left, A times the matrix with columns $\mathbf{v}_1, \mathbf{v}_2$ has columns $A\mathbf{v}_1, A\mathbf{v}_2$. On the right, the matrix with columns $\mathbf{u}_1, \mathbf{u}_2$ times a diagonal matrix scales column i by σ_i:

Left side: A times the matrix of the $\mathbf{v}$'s.

$$A \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix}$$

Right side: the matrix of the $\mathbf{u}$'s times the diagonal matrix.

$$\begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

The two sides are equal. In short:

$$AV = U\Sigma$$

Multiply both sides on the right by V^T. Since $VV^\mathsf{T} = I$, the V disappears from the left.

1. **In words:** every matrix is an **orthogonal matrix** (G-1407), times a diagonal matrix of non-negative numbers, times another orthogonal matrix.
2. **Formula:**
$$A = U\Sigma V^\mathsf{T}$$
The columns of U are the **left singular vectors** (G-1079) $\mathbf{u}_i$, the columns of V are the **right singular vectors** (G-1692) $\mathbf{v}_i$, and the diagonal of Σ holds the **singular values** (G-1812) σ_i.
3. **Example:**
$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$
The three factors:
$$U = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$$
$$\Sigma = \begin{bmatrix} 6.71 & 0 \\ 0 & 2.24 \end{bmatrix}$$
$$V^\mathsf{T} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$
Their product is A:
$$A = U\Sigma V^\mathsf{T}$$
Multiplying out with the exact singular values gives back A exactly:
$$\sigma_1 = 3\sqrt{5}$$
$$\sigma_2 = \sqrt{5}$$
Computing the SVD does it step by step.

4.2 Reading the three factors as moves

Key point: Read $U\Sigma V^\mathsf{T}\mathbf{x}$ from right to left: rotate, stretch, rotate.

A product of matrices is one transformation after another, applied from right to left (see why the product is written right to left). Changing to an eigenbasis reads PDP^{-1} this way: translate into the coordinates of the **eigenbasis** (G-664) with the **change of basis matrix** (G-374), scale each coordinate, translate back. $U\Sigma V^\mathsf{T}$ has the same "translate, transform, translate back" shape, with one difference: it uses two bases. V^T translates $\mathbf{x}$ into coordinates along $\mathbf{v}_1, \mathbf{v}_2$; Σ scales each coordinate; U translates the result out along $\mathbf{u}_1, \mathbf{u}_2$. Because both bases are **orthonormal** (G-1409), both translations are rotations (or flips).

Figure 1 follows the unit circle through the three moves:

1. V^T **rotates.** V^T is the **rotation** (G-1709) by $-45°$. It carries $\mathbf{v}_1$ onto $\hat{\imath}$ (written $\mathbf{e}_1$ in the figure) and $\mathbf{v}_2$ onto $\hat{\jmath}$ ($\mathbf{e}_2$). The circle looks the same, since a rotated circle is still a circle.
2. Σ **stretches.** Σ is a diagonal matrix, so it stretches along the axes only (as feature scaling did in feature scaling is a stretch along the axes): by 6.71 along x and 2.24 along y. The circle becomes an ellipse lying along the axes.
3. U **rotates.** U is the rotation by 71.6°. The rotation turns the ellipse so that its axes point along $\mathbf{u}_1$ and $\mathbf{u}_2$.

The final ellipse is exactly the one A makes in one step (the dashed red curve in the last frame).

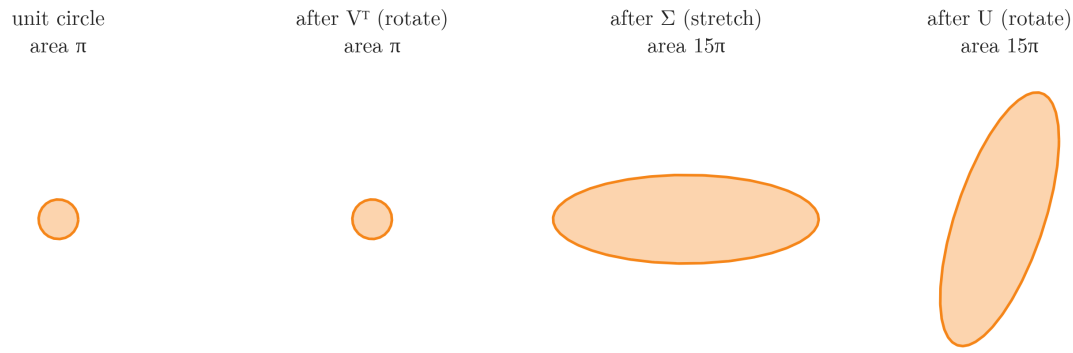


Figure 6: Area of the unit circle after each factor: the two rotations keep it at π, the stretch Σ multiplies it by $\sigma_1\sigma_2 = 15$

Figure 6 tracks the area through the three moves: only the middle step changes it, which is the area check in the Extra box below. Every **linear transformation** (G-1097) of the plane is this simple underneath: a rotation, a stretch along two perpendicular directions, and another rotation. Either rotation may also include a flip.

Extra: Each rotation preserves area, so all the area change happens in Σ. The area rule gives a check: $|\det A| = \sigma_1\sigma_2$, where the **determinant** $\det A$ (G-598) is the factor by

which A scales areas. Here the determinant of A is:

$$\det A = 3 \times 5 - 0 \times 4 = 15$$

The product of the **singular values** (G-1812) is the same:

$$\sigma_1 \sigma_2 = 3\sqrt{5} \times \sqrt{5} = 15$$

A singular value of 0 means Σ squishes one axis flat, which is the determinant-0 case (see when a matrix sends a non-zero vector to zero).

5. Singular values and singular vectors

Key point: Singular values are never negative and are listed largest first; the number of non-zero ones is the **rank** (G-1627); the largest is the most any unit vector can be stretched.

5.1 Ordering and sign

Key point: We sort $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$; a sign flip of a matching $\mathbf{u}_i$ and $\mathbf{v}_i$ gives an equally valid SVD.

- **Never negative.** A **singular value** (G-1812) is a length (of $A\mathbf{v}_i$), so $\sigma_i \geq 0$. If a stretch would need a negative number, we flip the sign of $\mathbf{u}_i$ instead.
- **Largest first.** By convention $\sigma_1 \geq \sigma_2 \geq \ldots$. Then Σ is unique for every matrix (MML Thm 4.22).
- **Signs of the vectors.** Changing both $\mathbf{u}_i$ and $\mathbf{v}_i$ to $-\mathbf{u}_i$ and $-\mathbf{v}_i$ keeps $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ true. So libraries may return vectors with signs opposite to ours, as NumPy does below. Flipping only one of the pair is wrong; computing the SVD shows what goes wrong.

5.2 Rank and the largest stretch

Key point: The rank is the number of non-zero **singular values** (G-1812); σ_1 is the longest $A\mathbf{x}$ can be for a unit vector $\mathbf{x}$.

- **Rank.** Each non-zero σ_i gives one independent output direction $\mathbf{u}_i$; a zero σ_i squishes its direction to nothing. So the **rank** (G-1627) of the matrix (see rank) is the number of non-zero singular values. A has two, so its rank is 2.
- **Largest stretch.** Every unit vector lands somewhere on the ellipse, and the farthest points of the ellipse are the ends of its long axis. So $\|A\mathbf{x}\| \leq \sigma_1$ for every unit $\mathbf{x}$, and σ_1 is reached at $\mathbf{x} = \mathbf{v}_1$. In the same way, σ_2 is the smallest stretch of a 2×2 matrix.

For A: $\hat{\imath}$ lands on $[3, 4]$, of length 5; $\hat{\jmath}$ on $[0, 5]$, also of length 5. Both are below $\sigma_1 = 6.71$ and above $\sigma_2 = 2.24$.

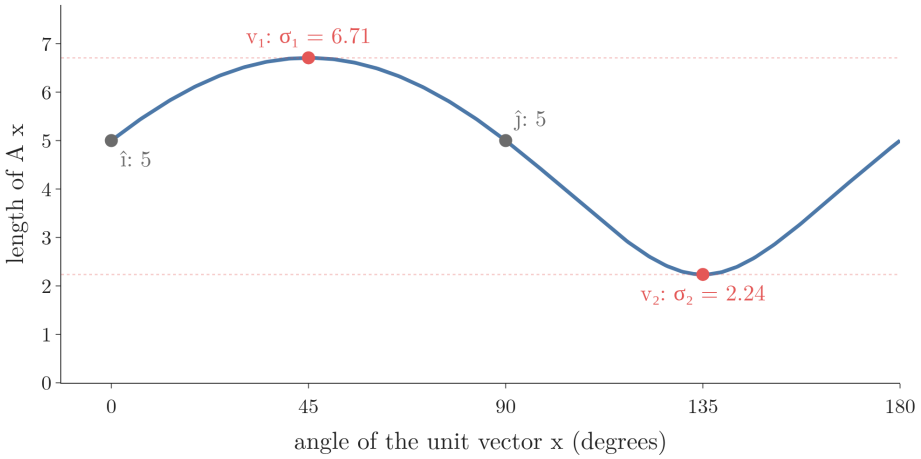


Figure 7: The length of $A\mathbf{x}$ for every unit vector $\mathbf{x}$, by its angle: the curve never leaves the band between σ_2 and σ_1

In Figure 7, watch the top and bottom of the curve: the top is $\sigma_1 = 6.71$ at 45° ($\mathbf{v}_1$), the bottom is $\sigma_2 = 2.24$ at 135° ($\mathbf{v}_2$), and $\hat{\imath}$ and $\hat{\jmath}$ sit in between at 5.

> **Python:** NumPy's `svd` returns U, the singular values as a 1D array, and V^T (called `Vh`).
>
> ```
> import numpy as np
>
> A = np.array([[3, 0], [4, 5]])
> U, s, Vt = np.linalg.svd(A)
> s      # [6.708, 2.236]
> U      # [[-0.316, -0.949], [-0.949, 0.316]]
> Vt     # [[-0.707, -0.707], [-0.707, 0.707]]
> U @ np.diag(s) @ Vt      # gives back A
> ```
>
> The third output is already V^T: its **rows** are the $\mathbf{v}_i$. NumPy's $\mathbf{u}_1$ and $\mathbf{v}_1$ are both the negatives of ours, an equally valid choice (Section 5.1).

6. Matrices that are not square

> **Key point:** For an $m \times n$ matrix, U is $m \times m$, Σ is $m \times n$ with zeros below (or beside) its diagonal, and V is $n \times n$. The **thin SVD** (G-1969) drops the parts that only multiply zeros.

6.1 The full SVD

> **Key point:** Σ has the same shape as A; its **singular values** (G-1812) sit on the diagonal, padded with zeros.

A matrix that is not square moves vectors between spaces of different dimensions. Its columns are still the landing spots of the basis vectors. A 3×2 matrix has two columns, so the input

has two basis vectors (2D), and each landing spot has three coordinates, so the output is 3D: the whole input plane lands on a flat plane through the origin of 3D space, its **column space** (G-414). A 2×3 matrix does the reverse and squashes 3D space onto a plane.

A data matrix is rarely square: it has many more rows (**observations** (G-1374), one per record) than columns (**features** (G-772), one per input variable). The SVD still exists. For an $m \times n$ matrix A:

- V is $n \times n$: an **orthonormal** (G-1409) basis of the input space $\mathbb{R}^n$ ($\mathbb{R}^n$ means all lists of n numbers, such as $(2, 3)$ in $\mathbb{R}^2$);
- U is $m \times m$: an orthonormal basis of the output space $\mathbb{R}^m$;
- Σ is $m \times n$, the same shape as A, with $\sigma_1, \sigma_2, \ldots$ on its diagonal and zeros everywhere else.

This factorisation is the **full SVD** (G-810) (Figure 9, top). Take the 3×2 matrix

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

B takes 2D vectors to 3D vectors. Its singular values are $\sqrt{3} \approx 1.73$ and 1, so

$$\Sigma = \begin{bmatrix} 1.73 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

with a row of zeros at the bottom. U is 3×3; its third column is a direction in 3D that no output of B ever reaches.

Geometrically the picture is the same as in 2D, with one more dimension in the output. V^{T} rotates the unit circle in the plane; Σ stretches it into an ellipse and places it in 3D space, on the flat floor where the third coordinate is 0; U turns that flat ellipse to its final tilt in 3D. The outputs of B fill only a tilted plane in 3D, never all of space.

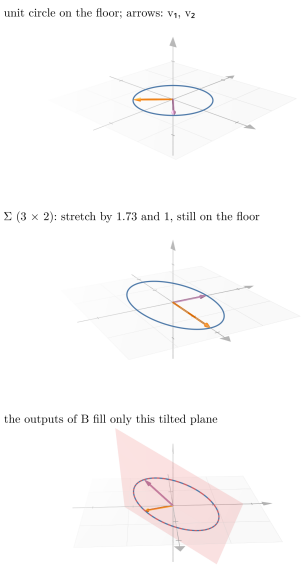


Figure 8: The SVD of the 3×2 matrix B in 3D. The unit circle lies on the floor (third coordinate 0) with $\mathbf{v}_1$ (orange) and $\mathbf{v}_2$ (purple). V^T turns it on the floor, Σ stretches it into an ellipse of axes 1.73 and 1, still on the floor, and U tilts it up onto the plane of B's outputs (red). The dashed curve is B applied in one step. 2D-to-3D picture after 3Blue1Brown, "Nonsquare matrices as transformations between dimensions"

In Figure 8, watch the floor: the first two moves never leave it, and only U, the 3×3 **rotation** (G-1709), lifts the ellipse into 3D. It lands exactly on the dashed curve, and the red plane is everything B can output.

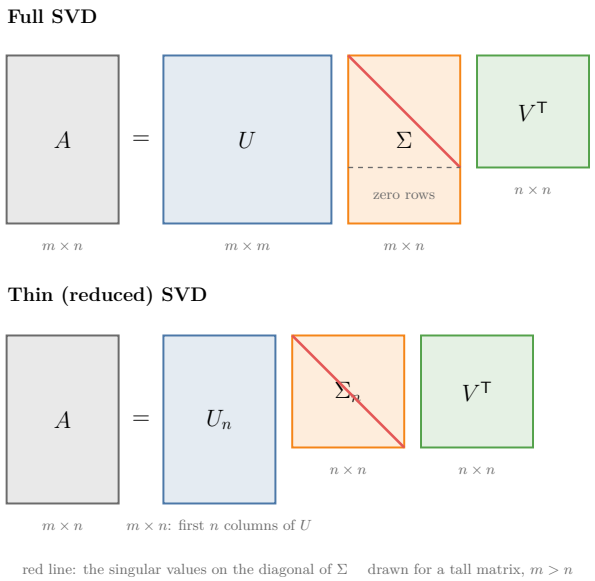


Figure 9: Shapes of the factors in the full and in the thin SVD of a tall matrix

6.2 The thin (reduced) SVD

> **Key point:** Keep only the first n columns of U and the square top of Σ; the product is unchanged.

In the **full SVD** (G-810), the zero rows of Σ multiply the last $m - n$ columns of U by zero. Those columns never affect A, so we can drop them. The result is the **thin SVD** (G-1969), also called the **reduced SVD** (MML §4.5; Figure 9, bottom):

$$A = U_n \Sigma_n V^\top$$

with these shapes:

- U_n is $m \times n$;
- Σ_n is $n \times n$;
- V is $n \times n$.

Now Σ_n is a square diagonal matrix, as in an eigen-decomposition. For data with 10,000 rows and 5 columns, the full U would be $10{,}000 \times 10{,}000$ (100 million numbers), while the thin U_n is $10{,}000 \times 5$. The thin SVD is what we use in practice.

> **Extra:** For a wide matrix ($m < n$) everything mirrors: Σ gets zero columns on the right, and the thin SVD keeps the first m columns of V. Some books go one step further and keep only the r columns that belong to non-zero **singular values** (G-1812), where r is the **rank** (G-1627); all three versions describe the same factorisation.

> **Python:** `full_matrices=False` gives the thin SVD.

```
B = np.array([[1, 1], [0, 1], [1, 0]])
U, s, Vt = np.linalg.svd(B)                          # full
U.shape, s, Vt.shape        # (3, 3), [1.732, 1.0], (2, 2)
Un, s, Vt = np.linalg.svd(B, full_matrices=False) # thin
Un.shape                    # (3, 2)
Un @ np.diag(s) @ Vt        # gives back B
```

7. SVD and eigen-decomposition compared

> **Key point:** The eigen-decomposition uses one basis that need not be perpendicular and exists only for some square matrices; the SVD uses two **orthonormal** (G-1409) bases and exists for every matrix.

Both factorisations have the same three-step shape: change of basis, scale each new axis, change basis again. Figure 5 shows the difference on A. Its eigenvectors, $[1, -2]$ (eigenvalue 3) and $[0, 1]$ (eigenvalue 5), each stay on their own line, but the two lines are not perpendicular. Its singular vectors are perpendicular and land on perpendicular vectors, but each one turns.

	Eigen-decomposition $A = PDP^{-1}$	SVD $A = U\Sigma V^{\mathsf{T}}$
Which matrices	square, with an eigenbasis	every matrix, any shape
Bases	one basis, P, used in and out	two: V for inputs, U for outputs
Perpendicular?	not in general	always orthonormal
Undoing the basis	P^{-1} must be computed	V^{T}, free
Diagonal entries	eigenvalues: any sign, can be complex	singular values: real, ≥ 0
For A	3 and 5	6.71 and 2.24

The two agree for one important family. A **symmetric matrix** (G-1932) equals its own **transpose** (G-2012). When it is also **positive semi-definite** (G-1532) (all its eigenvalues are 0 or positive), its eigenvectors are orthonormal and its eigenvalues are non-negative, so P is orthogonal and $A = PDP^{\mathsf{T}}$ is already an SVD, with $U = V = P$. A covariance matrix is exactly such a matrix (see the eigenvectors of the covariance matrix), so for it the eigenvalues and the **singular values** (G-1812) are the same numbers (MML §4.6).

> **Extra:** Even for a symmetric matrix the two can differ when an eigenvalue is negative. The matrix with rows $[0, 1]$ and $[1, 0]$ swaps x and y: its eigenvalues are 1 and -1, but its singular values are 1 and 1, because singular values ignore the flip and report only the stretch.

8. Summary

Object	Meaning	For A with rows $[3,0]$, $[4,5]$
$\mathbf{v}_1, \mathbf{v}_2$ (columns of V)	perpendicular input directions that stay perpendicular	$[0.707, 0.707]$, $[-0.707, 0.707]$
σ_1, σ_2 (diagonal of Σ)	stretch factors, largest first	6.71, 2.24
$\mathbf{u}_1, \mathbf{u}_2$ (columns of U)	axes of the output ellipse	$[0.316, 0.949]$, $[-0.949, 0.316]$
V^T, Σ, U	rotate, stretch, rotate	$-45°$; $\times 6.71$ and $\times 2.24$; $+71.6°$

- Every matrix sends the unit circle to an ellipse; $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ names its axes, so the longest and shortest stretch of any unit vector can be read off the ellipse.
- An **orthogonal matrix** (G-1407) has orthonormal columns, only rotates or flips, and has $Q^{-1} = Q^\mathsf{T}$, because $Q^\mathsf{T}Q = I$; so undoing it costs nothing, and it keeps every length and angle.
- $A = U\Sigma V^\mathsf{T}$ for every matrix: rotate, stretch along the axes, rotate, because the output direction $\mathbf{u}_i$ is allowed to differ from the input direction $\mathbf{v}_i$, unlike an eigenvector.
- **Singular values** (G-1812) are ≥ 0, because each is a length; their count above zero is the **rank** (G-1627), because a zero σ_i squishes its direction to nothing; σ_1 is the largest stretch.
- For tall data, the **thin SVD** (G-1969) keeps only the first n columns of U, because the other columns only multiply zeros of Σ; it saves memory and is what we use in practice.
- For a covariance matrix, the SVD and the eigen-decomposition coincide, because it is symmetric with eigenvalues 0 or above, so its eigenvectors are already **orthonormal** (G-1409).
- So every matrix, square or not, is a **rotation** (G-1709), a stretch by its singular values, and another rotation.

9. Sources

Built from

- 3Blue1Brown (Sanderson, G.), "Linear transformations and matrices | Chapter 3, Essence of linear algebra", YouTube, https://www.youtube.com/watch?v=kYB8IZa5AuE
- Khan Academy (Khan, S.), "Orthogonal matrices preserve angles and lengths", YouTube, https://www.youtube.com/watch?v=yDwIfYjKEeo
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Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Definition 3.8, Theorem 4.22, sections 4.5–4.6

(MML): the circle-to-ellipse picture, the factorisation, singular values and the thin SVD, which no video in the list covers.

- Strang, G. (2016). *Introduction to Linear Algebra*, 5th ed. Wellesley-Cambridge Press. Sections 7.2 (bases and matrices in the SVD) and 7.4 (the geometry of the SVD).

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Rotation (G-1709)	A transformation that turns space without stretching or flipping it (an orthogonal matrix with determinant $+1$).
Singular value decomposition (SVD) (G-1813)	A way to write any matrix, square or not, as three simple steps: turn, stretch by the singular values, turn ($A = U\Sigma V^{\mathsf{T}}$: orthogonal, diagonal, orthogonal; a factorisation). ML uses it for PCA and to compress data.
Singular value (σ_i) (G-1812)	A stretch factor of a matrix: the length of $A\mathbf{v}_i$; never negative, listed largest first.
Thin (reduced) SVD (G-1969)	A smaller form of the SVD that drops the columns of U multiplied by the zero rows of Σ; those columns never affect the matrix, so it gives the same A from smaller matrices.
Orthogonal matrix (G-1407)	A square matrix that only rotates or flips: its columns have length 1 and are all perpendicular (orthonormal), and its inverse is its transpose.
Orthonormal (G-1409)	Vectors of length 1 that are all perpendicular to each other.
Reflection (G-1651)	A transformation that mirrors space, without stretching it (an orthogonal transformation with determinant -1).
Singular value equation (G-1814)	The equation $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$: the matrix sends each input direction $\mathbf{v}_i$ to its output direction $\mathbf{u}_i$, stretched by the singular value σ_i; it is used to compute each $\mathbf{u}_i$.
Left singular vector ($\mathbf{u}_i$) (G-1079)	One of the output directions of the SVD: the unit vector that a right singular vector lands on, stretched by its singular value ($A\mathbf{v}_i = \sigma_i\mathbf{u}_i$); the columns of U.

Term	Meaning
Right singular vector ($\mathbf{v}_i$) (G-1692)	A column $\mathbf{v}_i$ of V in the SVD: a unit-length input direction that the matrix sends to σ_i times the matching left singular vector $\mathbf{u}_i$.
Full SVD (G-810)	The version of the SVD that keeps every column of U and V, so both are square and Σ has the same shape as A, with the singular values on its diagonal padded with zeros. It gives complete orthonormal bases for both the input and the output space.
Symmetric matrix (G-1932)	A square matrix equal to its own transpose, such as $X^\mathsf{T}X$ or a covariance matrix; it always has a full set of perpendicular eigenvectors.
Positive semi-definite (G-1532)	A symmetric matrix that never curves downward in any direction: its eigenvalues are all 0 or positive. For such a matrix the eigenvectors are orthonormal, so its eigen-decomposition and its SVD are the same.
Eigenbasis (G-664)	A basis made of eigenvectors of a matrix. Seen in this basis the matrix only scales each basis vector, so it becomes a diagonal matrix, which makes powers such as A^k easy to compute.
Linear transformation (G-1097)	A change of the whole plane by a matrix that keeps grid lines straight and evenly spaced, and the origin fixed; algebraically, $L(\mathbf{v}+\mathbf{w}) = L(\mathbf{v}) + L(\mathbf{w})$ and $L(c\mathbf{v}) = cL(\mathbf{v})$.
Orthogonal (G-1408)	At right angles (perpendicular); two non-zero vectors are orthogonal when their dot product is 0, which as data means they share nothing.
Identity matrix (G-915)	The matrix that leaves every vector unchanged.
Transpose (G-2012)	A matrix or vector with rows and columns swapped; it turns a column vector u into a row u^T, so $u^\mathsf{T}x$ is the dot product.
Inverse matrix (G-968)	The matrix that undoes another: their product is the identity matrix.

Term	Meaning
Change of basis matrix (G-374)	A matrix P that translates between two sets of axes: its columns are the new axes (basis vectors). $P^{-1}AP$ is the same transformation as A seen in the new axes; in an eigenbasis it only scales.
Determinant (G-598)	The factor by which a matrix scales areas (volumes in 3D); 0 when it squishes space into a lower dimension.
Rank (of a matrix) (G-1627)	The number of linearly independent columns of a matrix, that is, how many directions its columns really span; a column that is a combination of others does not add to it.
Column space (G-414)	Every output a matrix can produce: all the vectors made by scaling and adding its columns (the span of its columns).
Observation (G-1374)	One record of a dataset: one row of the data table.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).